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Distributed denial of service (DDoS) classification based on random forest model with backward elimination algorithm and grid search algorithm

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Distributed Denial of Service (DDoS) attacks pose significant threats to network security, disrupting critical services by overwhelming targeted systems with malicious traffic. In this study, a machine learning-based approach is proposed to classify DDoS attacks using multiple classification models, including Random Forest (RF), Naïve Bayes (NB), K-Nearest Neighbors (KNN), Linear Discriminant Analysis (LDA), and Support Vector Machine (SVM). The DDoS-SDN dataset was used for training and evaluation, with feature selection via Backward Elimination (BE) and hyperparameter tuning using Grid Search with 5-fold Cross-Validation (CV = 5). Experimental results demonstrate a significant improvement in classification performance after feature selection and parameter optimization, with RF achieving the highest accuracy of 99.99%. In this study, we propose a machine learning-based classification framework enhanced by feature selection and hyperparameter optimization techniques through employing Recursive Feature Elimination (RFE) and Grid Search .Our model based on Random Forest (RF) achieved a remarkable accuracy of 99.99%, outperforming other baseline classifiers, including Naive Bayes (98.85%), K-Nearest Neighbors (97.90%), Linear Discriminant Analysis (97.10%), and Support Vector Machine (95.70%). In addition to accuracy, the RF model also demonstrated superior F1 score, recall, and precision, each reaching 99.99%. These results validate the effectiveness of our optimization strategy in improving classification performance. The study highlights the effectiveness of feature engineering and model optimization in enhancing DDoS detection accuracy, making machine learning a viable solution for real-time cybersecurity applications.

Keywords DDoS attack, Random forest, Feature selection, Hyperparameter tuning, Grid search, Network security, Cybersecurity, Classification models

Internet of Things (IoT) and similar networking technologies continue to experience consistent growth regarding their user base, infrastructure components (such as networking equipment), and software applications. Current data indicates that the quantity of devices connected to these networks will maintain its upward trajectory in the future[1]. The ongoing push toward uninterrupted and decentralized connectivity, applications hosted in the cloud, and automated real-time network monitoring suggests these developments will persist into the future. To meet rising user expectations and service quality requirements, academic researchers and industry professionals are collaboratively developing innovative approaches to network architecture design[2]. Additionally, safeguarding personal information and ensuring cloud system security have become increasingly vital concerns, as both the frequency and variety of attacks targeting cloud infrastructure continue to escalate[3]. One significant focus in this field is software-defined networks (SDNs), which separate control planes from data planes[4]. This key

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characteristic has resulted in numerous cutting-edge IoT network frameworks, including 5G and post-5G architectures, being built on SDN principles[5]. As a rapidly advancing technology, SDN streamlines network administration, management, and oversight. Through its controller, SDN provides network administrators with comprehensive network visibility while enabling interactive programming of underlying data plane components. The SDN controller functions as the network's central intelligence, directly managing data plane devices via application programming interfaces (APIs)[6]. However, despite SDN's programmability, flexibility, and distributed control—features crucial for successful implementation—these same attributes often conflict with security objectives[7]. As a result, many SDN architectures remain vulnerable to various security threats, including but not limited to heightened risk of Distributed Denial of Service (DDoS) flooding attacks[8]. DDoS attacks are one of threats facing network systems which disrupt applications and devices dependent on these networks. DDoS attacks operate by coordinating multiple systems against a single target, blocking legitimate users from accessing services including emails and websites. These attacks represent the most prevalent and severe threat both targeting and originating from Internet of Things (IoT) devices, Cloud Computing infrastructure, and fifth-generation (5G) communication networks[9]. DDoS flooding attacks are coordinated offensive operations that employ multiple compromised systems to deplete available network bandwidth or completely incapacitate target nodes or network-connected end-user equipment[10]. These attacks constitute the most hazardous type of malicious internet traffic. The perpetrators of DDoS flooding attacks typically utilize networks of infected devices (botnets) covertly to conduct their attacks. As a result, users whose devices are part of the compromised network are often oblivious to the fact that attacks are being launched from their equipment and IP addresses. Since the internet architecture lacks flow control requirements beyond end hosts, IoT devices are particularly susceptible to synchronized attacks like DDoS flooding[11]. DDoS attacks are typically categorized into two main groups: infrastructure layer attacks and application layer attacks. Infrastructure layer attacks comprise volume-based attacks and protocol attacks. Volume-based attacks focus on overwhelming the victim's bandwidth with massive traffic volumes. These attacks require minimal technical expertise, making them the most prevalent form of DDoS attack. Protocol attacks concentrate on depleting resources from the victim's server or networking equipment. Application layer attacks target web server functionality, aiming to disable it completely. As DDoS attack technologies continue to evolve, researchers work diligently to enhance mitigation strategies. Most current DDoS mitigation frameworks implement centralized solutions deployed at the host location, in cloud environments, or at data scrubbing centers. While these centralized approaches can counter DDoS attacks, they often lack efficiency in terms of time and resource utilization[12]. In response to these challenges, researchers have increasingly focused on implementing advanced machine learning (ML) models to identify DDoS traffic patterns. ML offers sophisticated tools and methodologies for predicting cyber-attacks through rapid, automated processes[13].

The study presents an optimized Random Forest model using Backward Elimination and Grid Search, delivering strong accuracy. However, since many studies have employed similar methods, it's important to highlight what sets this approach apart. The study achieves high accuracy in classification, but real-world deployment comes with extra challenges like latency, scalability, and detecting zero-day attacks. To make the model more practical, testing it on real network traffic or assessing its performance against adversarial threats would be valuable. While many studies have used machine learning for DDoS detection, this research takes it a step further with an improved Random Forest model, featuring optimized feature selection and hyperparameter tuning. Given the near-perfect accuracy, we conduct a thorough overfitting analysis using cross-validation and learning curves to ensure the model generalizes well. Additionally, we examine how class imbalance affects performance and compare our approach with other tuning methods like Bayesian Optimization and Random Search. Random Forest is a powerful ensemble method that improves accuracy by combining multiple decision trees, which helps reduce variance. However, if the model achieves an almost perfect score, it might be overfitting. This could happen if the dataset has redundant features or isn't diverse enough, causing the model to memorize patterns rather than generalize. To check for overfitting, it's best to validate the model on unseen data or use cross-validation with different folds.

Distributed Denial-of-Service (DDoS) attacks are one of the most important threats to network infrastructure, and capable of overwhelming services and causing severe disruptions. Traditional DDoS detection methods, such as signature-based and rule-based systems often struggle with high false positive rates, limited adaptability to new attack vectors, and inadequate scalability in high-traffic environments. These limitations underscore the need for more intelligent and adaptive detection mechanisms. Recent advancements in machine learning (ML) have opened new areas for detecting complex and previously unseen attack patterns. The integration of feature selection techniques such as Recursive Feature Elimination (RFE) and hyperparameter tuning strategies like Grid Search or Randomized Search has shown promise in enhancing model accuracy, reducing overfitting, and improving generalization. By using these techniques, it is possible to build more robust and efficient DDoS detection models. In this study, we propose a machine learning-based DDoS detection framework that incorporates both feature selection and hyperparameter optimization to address the shortcomings of traditional models and achieve superior performance across various evaluation metrics.

Related works

Recent intrusion detection research has extensively employed machine learning techniques. As machine learning algorithms continue to advance, more sophisticated intrusion detection systems are being developed. This section examines current literature regarding DDoS attacks specifically targeting cloud-based environments. Random Forest has emerged as a particularly effective algorithm; achieving over 96% accuracy in classifying DDoS attacks[14]. Feature selection methods, such as Recursive Feature Elimination, have been shown to enhance detection efficiency while maintaining high performance, with one study reporting 99.9% accuracy, precision, recall, and F1score[15]. Other approaches have combined Random Forest with additional techniques, such as C4.5

and GridSearchCV, to optimize model parameters[16]. Comparative studies have also evaluated the performance of Random Forest against other algorithms like Logistic Regression and K-Nearest Neighbor[17]. These studies collectively demonstrate the potential of machine learning, particularly Random Forest, in improving DDoS detection systems, with feature selection playing a crucial role in enhancing efficiency and accuracy.

Recent advancements in DDoS detection have explored the use of deep learning and hybrid models to improve accuracy and adaptability[18–25]. Deep learning applications in IoT-based security and bioinformatics, emphasizing their role in real-time data processing and pattern recognition[26–30]. Several studies have demonstrated how nature-inspired algorithms and cloud-based optimizations can enhance detection capabilities in large-scale systems[31–33].

Research proposes a Random Forest machine learning model for identifying Distributed Denial of Service (DDoS) attacks with substantial precision[34]. An investigation documented in[35] assessed the efficacy of diverse machine learning classification algorithms including Logistic Regression, Decision Tree, Random Forest, Ada Boost, Gradient Boost, KNN, and Naive Bayes—for detecting DDoS attacks utilizing the CIC-DDoS2019 dataset. This study determined that AdaBoost and Gradient Boost algorithms deliver superior classification outcomes, while Logistic Regression, KNN, and Naive Bayes yield satisfactory classification results, and Decision Tree and Random Forest generate inferior classification outcomes. Research presented in[36] examined the effectiveness of the Random Forest machine learning algorithm for identifying DDoS attacks, concluding that it outperforms other compared algorithms regarding accuracy and computational efficiency. The analysis in[37] introduced a methodology combining random forest and correlation-based feature selection (CFS) to categorize DDoS attacks with high precision and accuracy rates.

The author in[38] demonstrated the use of machine learning approaches to detect and classify four types of distributed denial of service (DDoS) attacks using the CICDDoS-2019 and UNSW-NB15 datasets. The author in[39] used machine learning techniques, specifically Random Forest and XGBoost classification algorithms, to classify and predict DDoS attack types using the UNWS-np-15 dataset, achieving high precision, recall, and accuracy with the XGBoost algorithm performing slightly better than the Random Forest algorithm. Two new Random Forest models are presented in[40]: one that can detect DDoS attacks from normal IP traffic with 99.99% accuracy and another that can classify 10 types of DDoS attacks with 99.92–99.99% accuracy, with processing times that allow for real-world application. The study in[41] presented a machine learning-based approach to detect DDoS attacks, with the random forest classification technique achieving the highest accuracy of 98.70%. The study in[42] evaluated and ranked supervised machine learning algorithms for their performance in classifying and detecting Distributed Denial of Service (DDoS) attacks, with the goal of reducing errors and improving precision and recall while maintaining detection accuracy. The author in[43] incorporated three well-known classification techniques: Multilayer Perceptron (MLP), Naive Bayes and Random Forest. The experimental results show that MLP achieved the highest accuracy rate (98.63%).

A supervised machine learning approach is introduced in[44] using Random Forest, KNN, and Logistic Regression to classify DDoS attacks, with Random Forest achieving the highest accuracy. A machine learning-based method is proposed in[45] for detecting DDoS attacks in cloud computing using feature selection and Random Forest. An intrusion detection model is proposed[46] using RBF-SVM with grid search to detect DDoS attacks in SDN-based VANETs. A hybrid deep learning approach is proposed in[47] that combines autoencoder and multilayer perceptron for effective DDoS attack detection and classification. The study in[48] evaluated the performance of various machine learning models for detecting distributed denial of service (DDoS) attacks using feature selection and hyper-parameter optimization techniques.

To improve model performance, hyperparameter tuning was performed using Grid Search[33,49–53]. Optimization strategies such as nature-inspired algorithms and metaheuristics have been widely adopted for similar tasks[54–56]. The effectiveness of such methods in improving prediction and classification accuracy has been demonstrated in various fields including healthcare, IoT, and sustainable agriculture.

Materials and methods

In this study, we propose a machine learning-based classification framework enhanced by feature selection and hyperparameter optimization techniques through employing Recursive Feature Elimination (RFE) and Grid Search. Our models are based on Random Forest, Naive Bayes, K-Nearest Neighbors, Linear Discriminant Analysis, and Support Vector Machine. The selected models, Random Forest, Naïve Bayes, K-Nearest Neighbors (KNN), Linear Discriminant Analysis (LDA), and Support Vector Machine (SVM) were chosen because of diversity in learning paradigms and decision boundaries, balance between accuracy, interpretability, and computational cost, and suitable for tabular data and binary/multiclass classification tasks typical of DDoS datasets.

Dataset description

This study used DDoS SDN dataset, available at https://www.kaggle.com/datasets/aikenkazin/ddos-sdn-dataset that contains 104,345 rows and 23 columns. There is a one target variable called label: contains only 1(malicious) and 0(benign). Our task is to classify whether the traffic is normal or not using classical Machine Learning algorithms. Table 1.displays the statistical analysis for the dataset features. Figure 1 show the heatmap matrix for the features. Our dataset analysis revealed a significant class imbalance, with far more benign traffic than DDoS attack instances. To address this, we used SMOTE to generate synthetic attack samples and also experimented with random under sampling to balance the dataset. We then evaluated how these techniques affected classification performance. To ensure the model's reliability in detecting minority-class attacks, we measured its performance using Precision-Recall AUC.

	count	mean	std	min	50%	max
dt	10000.0	1.105728e+04	7.907925e+02	9756	1.139500e+04	1.226500e+04
switch	10000.0	2.481600e+00	8.077912e-01	1.0	3.000000e+00	4.000000e+00
pktcount	10000.0	7.529119e+04	4.036941e+04	284	7.674100e+04	1.350360e+05
bytecount	10000.0	7.966358e+07	4.279166e+07	3027	8.180591e+07	1.439484e+08
dur	10000.0	1.952510e+02	1.143749e+02	0.0	1.920000e+02	4.730000e+02
dur_nsec	10000.0	5.513800e+08	2.351517e+08	790	5.590000e+08	9.140000e+08
tot_dur	10000.0	1.958208e+11	1.143912e+11	837	1.930000e+11	4.730000e+11
flows	10000.0	3.552100e+00	1.363991e+00	2.0	3.000000e+00	8.000000e+00
packetins	10000.0	1.912668e+03	2.442305e+02	558.0	1.943000e+03	2.242000e+03
pktperflow	10000.0	1.055789e+04	3.928351e+03	0.0	1.330800e+04	1.368400e+04
byteperflow	10000.0	1.118856e+07	4.220789e+06	0.0	1.418633e+07	1.458714e+07
pktrate	10000.0	3.515093e+02	1.308826e+02	0.0	4.430000e+02	4.560000e+02
port_no	10000.0	2.374900e+00	1.088792e+00	1.0	2.000000e+00	4.000000e+00
tx_bytes	10000.0	4.700992e+07	1.463899e+08	2845.0	3.826000e+03	1.269982e+09
rx_bytes	10000.0	4.699641e+07	1.075180e+08	926.0	3.539000e+03	9.905962e+08
tx_kbps	10000.0	9.025502e+02	2.903783e+03	0.0	0.000000e+00	2.058000e+04
rx_kbps	10000.0	9.027924e+02	2.305851e+03	0.0	0.000000e+00	1.657700e+04
tot_kbps	10000.0	1.805343e+03	3.481251e+03	0.0	0.000000e+00	2.058000e+04
label	10000.0	4.444000e-01	4.969239e-01	0.0	0.000000e+00	1.000000e+00

Table 1. Statistical analysis for the features.

Dataset collection and preprocessing

The first step in the methodology involves obtaining a dataset that contains network traffic records, including normal and DDoS attack instances. This dataset serves as the foundation for training and evaluating the machine learning models.

Data normalization

The collected dataset is subjected to data normalization, a preprocessing step that ensures uniformity across all features. Normalization scales the numerical values to a standard range (such as [0,1] or [-1,1]) to prevent features with larger magnitudes from dominating those with smaller magnitudes. This step enhances the performance and convergence speed of machine learning models.

Feature selection (BE—backward elimination)

Feature selection is applied to reduce dimensionality by selecting only the most relevant features. In this methodology, Backward Elimination (BE) is used. BE is a stepwise regression method that starts with all features and iteratively removes the least significant ones based on statistical criteria (e.g., p-values or feature importance). This process enhances model interpretability and reduces computational complexity while maintaining high classification accuracy. Backward Elimination (BE) was probably used because it systematically removes less important features, helping to reduce dimensionality while keeping performance high. However, other techniques like Recursive Feature Elimination (RFE) or SHAP-based feature selection could offer deeper insights and alternative perspectives.

Backward Elimination is used as feature selection technique to reduce the dimensionality of the dataset and to identify the most relevant features for DDoS attack classification. The process began with the full set of 23 features. At each iteration, a classification model was trained, and the p-values for each feature were calculated. Features with a p-value greater than 0.05 were removed, indicating they were not statistically significant for predicting the target feature. This iterative removal continues until all remaining features had p-values below 0.05, or the model's performance measured by accuracy, and no longer improved after removing a feature. A comparison of model performance before and after feature selection is presented in the results section, demonstrating the impact of Backward Elimination on the classification accuracy, precision, recall, and F1-score.

Hyperparameter optimization using grid search with Cross-Validation

Grid Search is great for hyperparameter tuning, but it can be quite resource intensive. To save time and computational power, alternatives like Bayesian Optimization or Random Search might be worth considering. Grid Search was our initial choice for hyperparameter tuning, but its high computational cost led us to explore other options. We compare Bayesian Optimization, which adjusts hyperparameters based on past results, with Random Search, which covers a wider range of possibilities. These methods are evaluated based on accuracy, tuning time, and resource usage to determine the most efficient approach. To fine-tune the models, a Grid Search technique is employed with 5-fold cross-validation (CV = 5). The steps involved in this process are:

- A predefined set of hyperparameters is tested for each model.

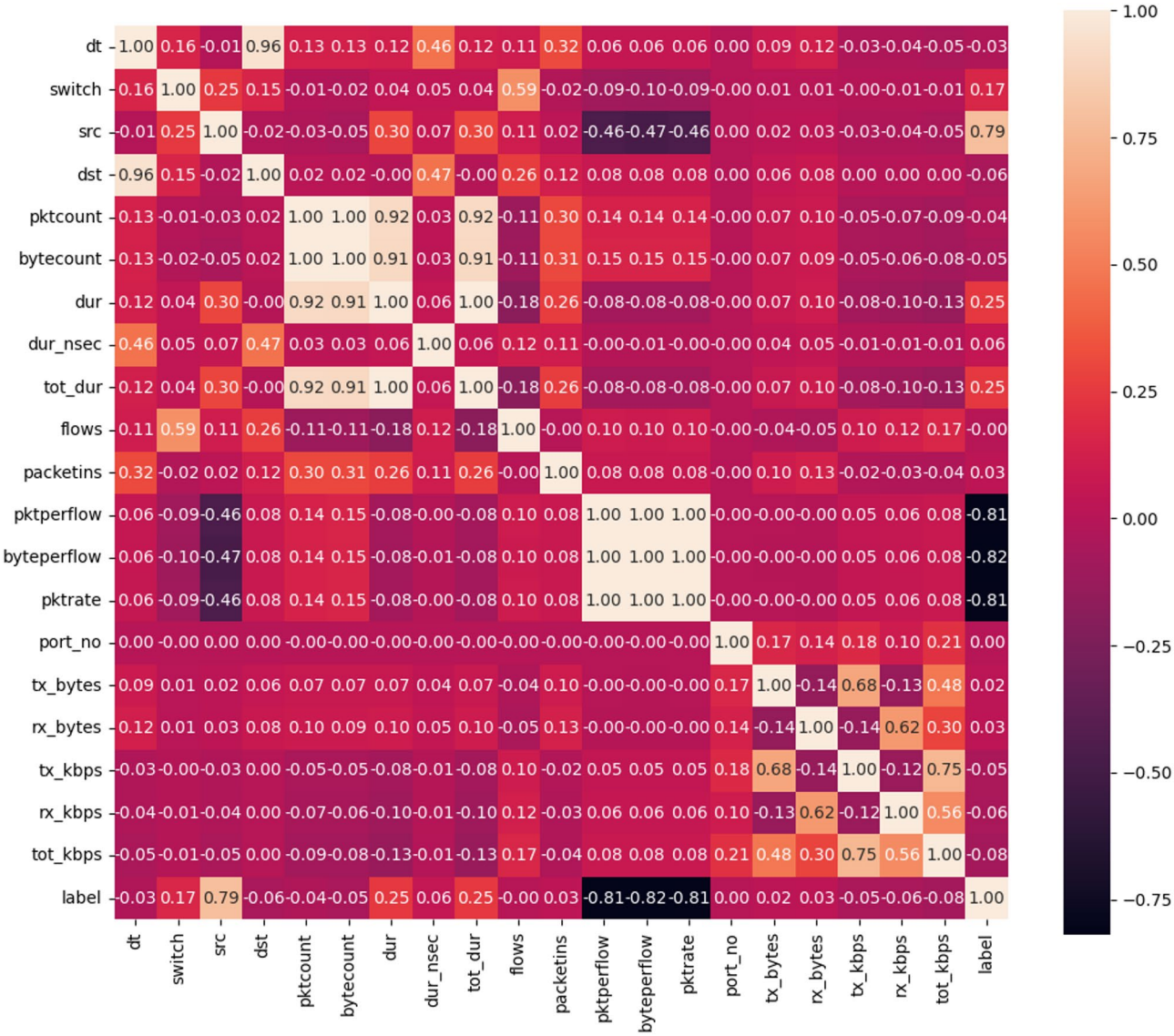

Fig. 1. Heatmap matrix for the features.

- The dataset is divided into 5 subsets, where the model is trained on 4 subsets and validated on the remaining one in a rotating manner.
- The combination of hyperparameters that yields the best performance is selected.

DDoS classification

Once the best hyperparameters are selected, the models are trained and evaluated on the dataset for DDoS attack classification. The study employs five different models (unspecified in the diagram but commonly used models include Decision Trees, Random Forest, Support Vector Machines, Neural Networks, and Gradient Boosting methods). Each model is evaluated based on various performance metrics such as accuracy, precision, recall, and F1-score to determine its effectiveness in detecting DDoS attacks.After training, tuning, and evaluating the models, the best-performing model is used for final predictions on new, unseen data. This step ensures that the system can detect and classify DDoS attacks in real-world scenarios with high reliability.

Results and discussion

Table 2. Presents the optimal hyperparameters obtained through Grid Search with 5-fold Cross-Validation[57] (CV = 5). Each model was tuned to enhance its classification performance. For Random Forest (RF), increasing the number of estimators to 200 and using entropy as the splitting criterion helped improve decision boundaries. Naïve Bayes (NB) benefited from adjusting the smoothing parameter (alpha = 0.2), which improved probabilistic estimation. K-Nearest Neighbors (KNN) performed best with 10 neighbors and uniform weighting, ensuring a balance between bias and variance. Linear Discriminant Analysis (LDA) retained three discriminant components to maximize class separation. Support Vector Machine (SVM) achieved optimal results using a

Models	Best hyperparameters
RF	N_estimators = 200, criterion = entropy.
NB	Alpha = 0.2.
KNN	N_neighbors = 10, weights = uniform.
LDA	N_components = 3.
SVM	Kernel = poly, C = 0.1.

Table 2. Best hyperparameters for the classification models using grid search algorithm.

Models	Accuracy	F1 score	Recall	Precision
RF	94.75%	94.75%	94.74%	94.75%
NB	93.92%	93.92%	93.91%	93.92%
KNN	91.84%	91.83%	91.84%	91.84%
LDA	91.37%	91.37%	91.37%	91.38%
SVM	90.03%	90.02%	90.03%	90.03%

Table 3. Performance for the classification models without using the selected features and without hyperparameters tuning.

Models	Accuracy	F1 score	Recall	Precision
RF	99.99%	99.99%	99.99%	99.99%
NB	98.85%	98.85%	98.85%	98.88%
KNN	97.90%	97.90%	97.90%	97.93%
LDA	97.10%	97.09%	97.10%	97.10%
SVM	95.70%	95.70%	95.70%	95.70%

Table 4. Performance for the classification models using the selected features and hyperparameters tuning.

polynomial kernel and a regularization parameter of C = 0.1, preventing overfitting while maintaining flexibility in decision boundaries. These selected hyperparameters were later used to train the models, leading to substantial performance improvements.

Table 3. provides the initial classification performance of the five models before applying feature selection and hyperparameter tuning. The Random Forest model achieved the highest accuracy of 94.75%, indicating that ensemble learning performed better than individual classifiers in the default configuration. Naïve Bayes followed closely with 93.92% accuracy, demonstrating the effectiveness of probabilistic classification. KNN and LDA achieved accuracies of 91.84% and 91.37%, respectively, showing moderate performance. SVM had the lowest accuracy (90.03%), likely due to the default kernel and regularization settings. These results highlight the need for feature selection and hyperparameter optimization to enhance model performance further.

Table 4. presents the classification results after applying Backward Elimination (BE) for feature selection and Grid Search for hyperparameter tuning. The results demonstrate a significant improvement across all models, validating the effectiveness of these optimization techniques. Random Forest exhibited the most remarkable improvement, increasing its accuracy from 94.75 to 99.99%, making it the best-performing model. Naïve Bayes and KNN showed substantial gains, achieving 98.85% and 97.90% accuracy, respectively, confirming that optimal parameter selection enhances classification effectiveness. LDA and SVM also improved, with final accuracies of 97.10% and 95.70%, respectively, indicating that both feature selection and hyperparameter tuning positively impacted their performance. These results highlight the critical role of data preprocessing and parameter tuning in optimizing DDoS detection models.

Figure 2 presents the confusion matrix for the Random Forest (RF) model, which achieved the highest classification accuracy of 99.99%. The matrix reveals minimal false positives and false negatives, demonstrating that RF accurately distinguishes between normal traffic and DDoS attacks. The model's ability to learn complex feature interactions and generalize well to unseen data makes it the most reliable classifier in this study. This result indicates that ensemble learning techniques are highly effective for DDoS detection.

Figure 3 illustrates the confusion matrix for the K-Nearest Neighbors (KNN) model, which achieved an accuracy of 97.90% after hyperparameter tuning. The matrix shows a noticeable reduction in misclassifications compared to the pre-tuning results, confirming that selecting an appropriate number of neighbors (N_neighbors = 10) improved classification. Despite this improvement, KNN remains slightly less effective than RF due to its sensitivity to noisy data and high-dimensional feature spaces.

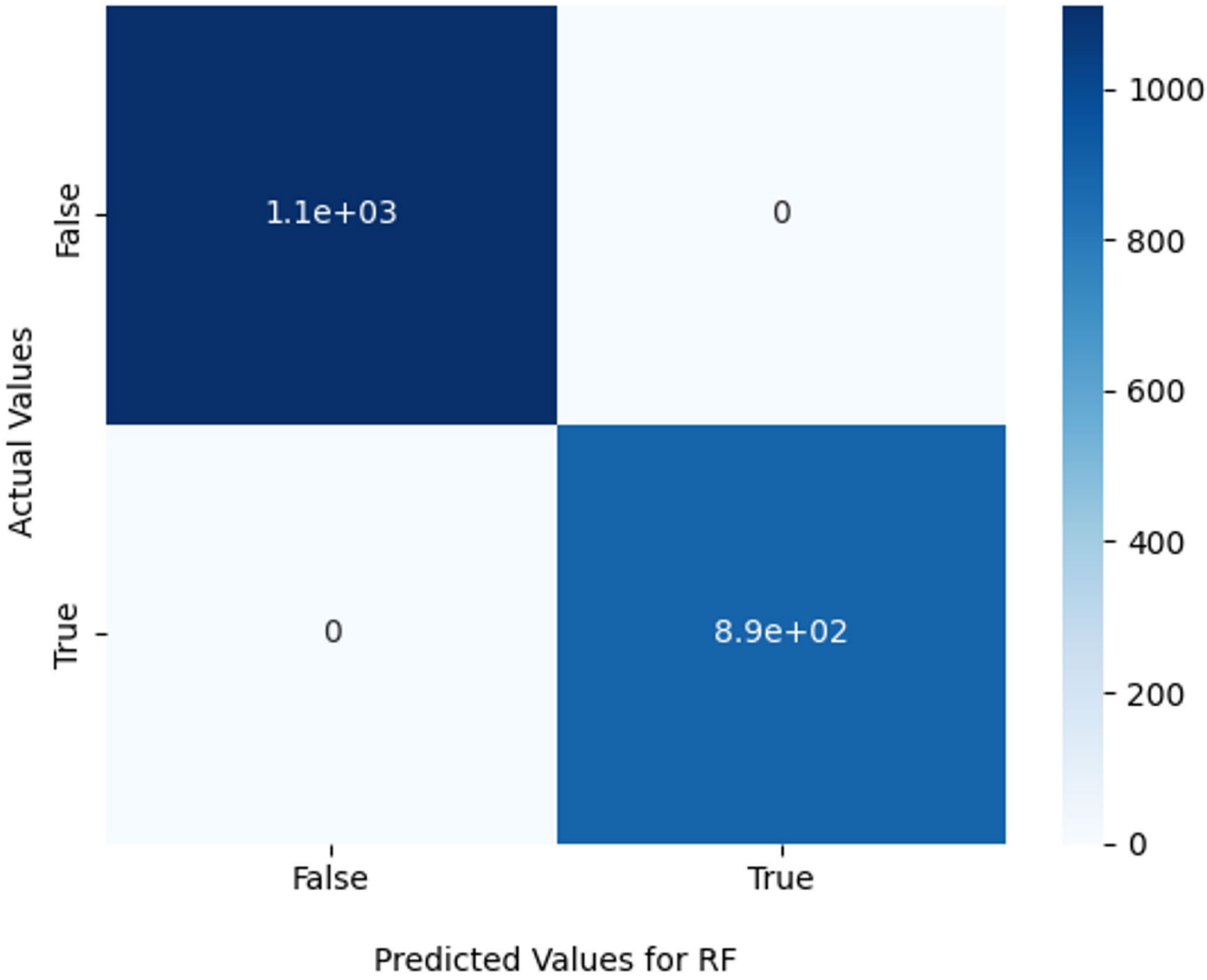

Fig. 2. Confusion matrix for RF model.

Figure 4 presents the confusion matrix for Naïve Bayes (NB), which reached an accuracy of 98.85% after tuning. The matrix indicates that NB exhibits high recall, meaning it correctly identifies most DDoS attacks with very few false negatives. This suggests that NB is particularly useful when minimizing false negatives is a priority, making it suitable for real-time anomaly detection. However, some misclassifications are still present, likely due to the assumption of feature independence, which may not hold for all traffic patterns.

Figure 5 shows the confusion matrix for Linear Discriminant Analysis (LDA), which achieved 97.10% accuracy after feature selection and parameter tuning. The matrix reveals that misclassifications decreased significantly compared to the pre-tuning results, validating the effectiveness of dimensionality reduction. LDA performed well in distinguishing between normal and attack traffic but remains slightly less effective than RF and NB. This suggests that LDA is a viable option when computational efficiency is a concern, as it balances performance with reduced complexity.

Figure 6 illustrates the confusion matrix for the Support Vector Machine (SVM) model, which improved from 90.03 to 95.70% accuracy after hyperparameter tuning. The matrix highlights a significant reduction in false positives and false negatives, demonstrating the impact of selecting an appropriate kernel function (poly) and regularization parameter ($C = 0.1$). While SVM showed notable improvement, it still had the lowest accuracy among the models, suggesting that further optimizations, such as using different kernel functions or feature transformations, could enhance its performance.

Figures (7,8, 9, 10, and 11) displays the ROC curves for the models used in this study.

Figure 12 represents the feature importance plot.

Table 5 demonstrates a comparison between grid search (GS) and random search (RS) methods.

The Wilcoxon signed rank test results are demonstrated in Table 6 that confirm the statistical significance of the results across algorithms, with all comparisons yielding a P-value of 0.003125 (two-tailed), indicating strong evidence against the null hypothesis.

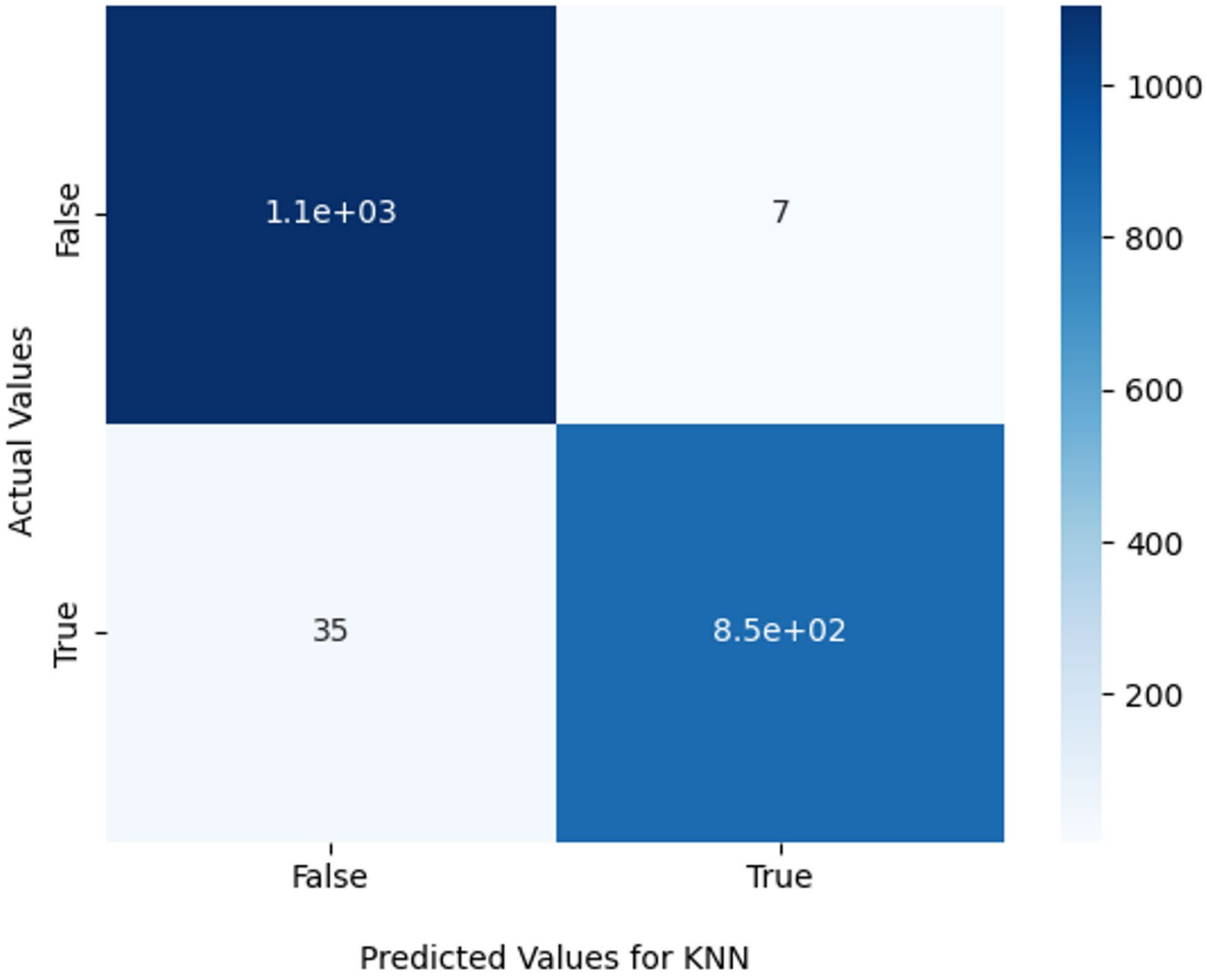

Fig. 3. Confusion matrix for KNN model.

Conclusion and future work

This study proposed a machine learning-based approach for DDoS attack detection using multiple classification models, emphasizing the importance of feature selection and hyperparameter tuning in improving classification accuracy. The results indicate that Random Forest outperformed all other models, achieving 99.99% accuracy, while Naïve Bayes, KNN, LDA, and SVM also demonstrated significant improvements after optimization. The findings suggest that feature selection via Backward Elimination (BE) and hyperparameter tuning with Grid Search (CV = 5) significantly enhance classification performance. The near-perfect accuracy achieved by RF highlights its suitability for real-time DDoS detection in modern network environments, including Software-Defined Networks (SDNs). While these results are good and deploying such a model in real-time network environments presents practical challenges. These include handling high-throughput traffic with minimal latency which maintain adaptability to new and evolving attack patterns, and managing the computational overhead associated with continuous monitoring. Future work could focus on the integration of deep learning techniques, such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), which may offer improved scalability and adaptability. Also, the use of ensemble learning strategies combining multiple classifiers could further enhance robustness and detection reliability. These directions aim to bridge the gap between experimental success and real-world deployment, making the system more resilient to diverse and dynamic network threats.

We plan to explore the performance of our trained models on datasets that include variations of the attack patterns present in the DDoS-SDN dataset. This could involve either augmenting the existing dataset with synthetic variations or acquiring datasets that contain diverse attack strategies. We aim to evaluate our models on completely unseen DDoS datasets. This would provide a more rigorous assessment of their ability to generalize new and potentially different attack scenarios.

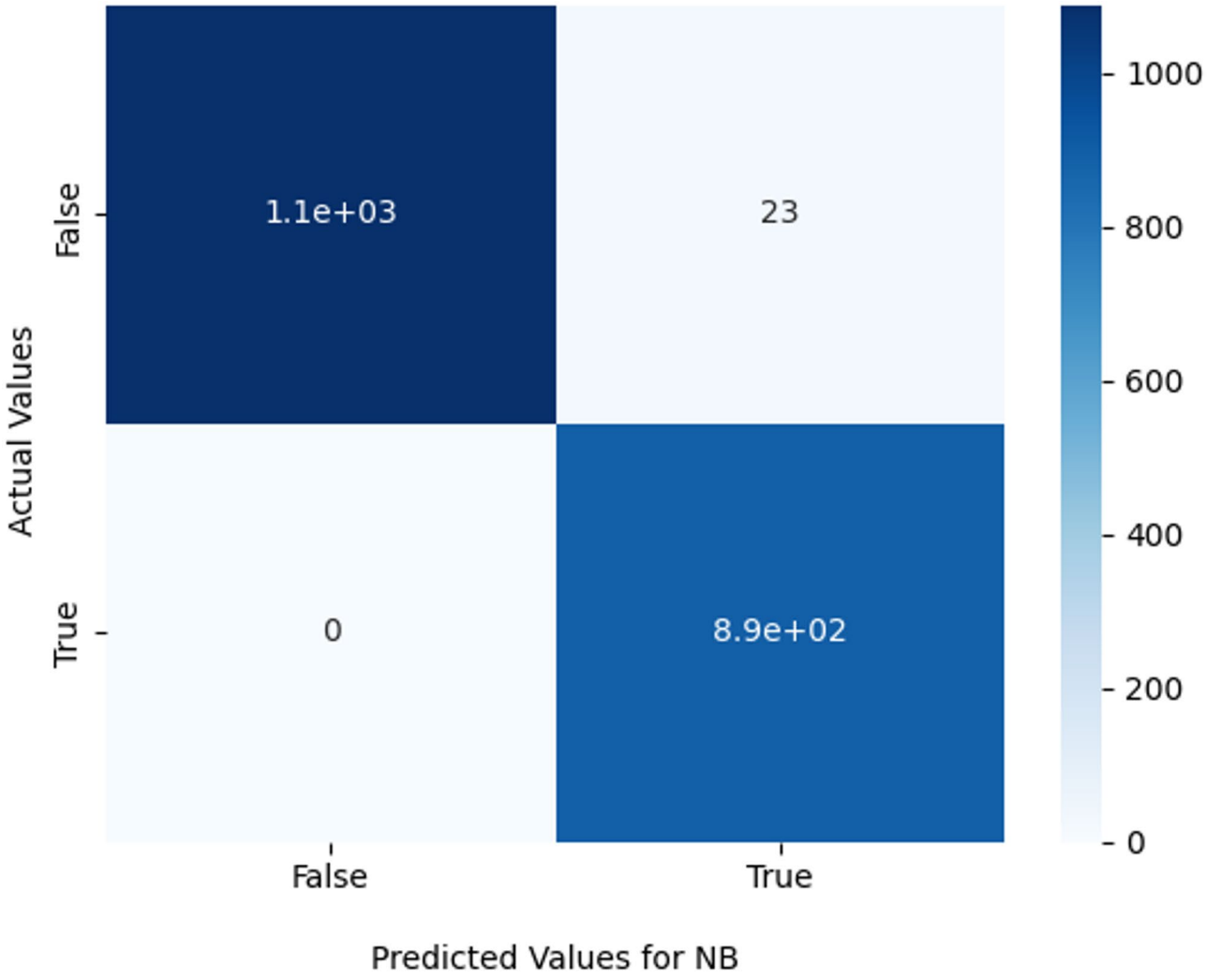

Fig. 4. Confusion matrix for NB model.

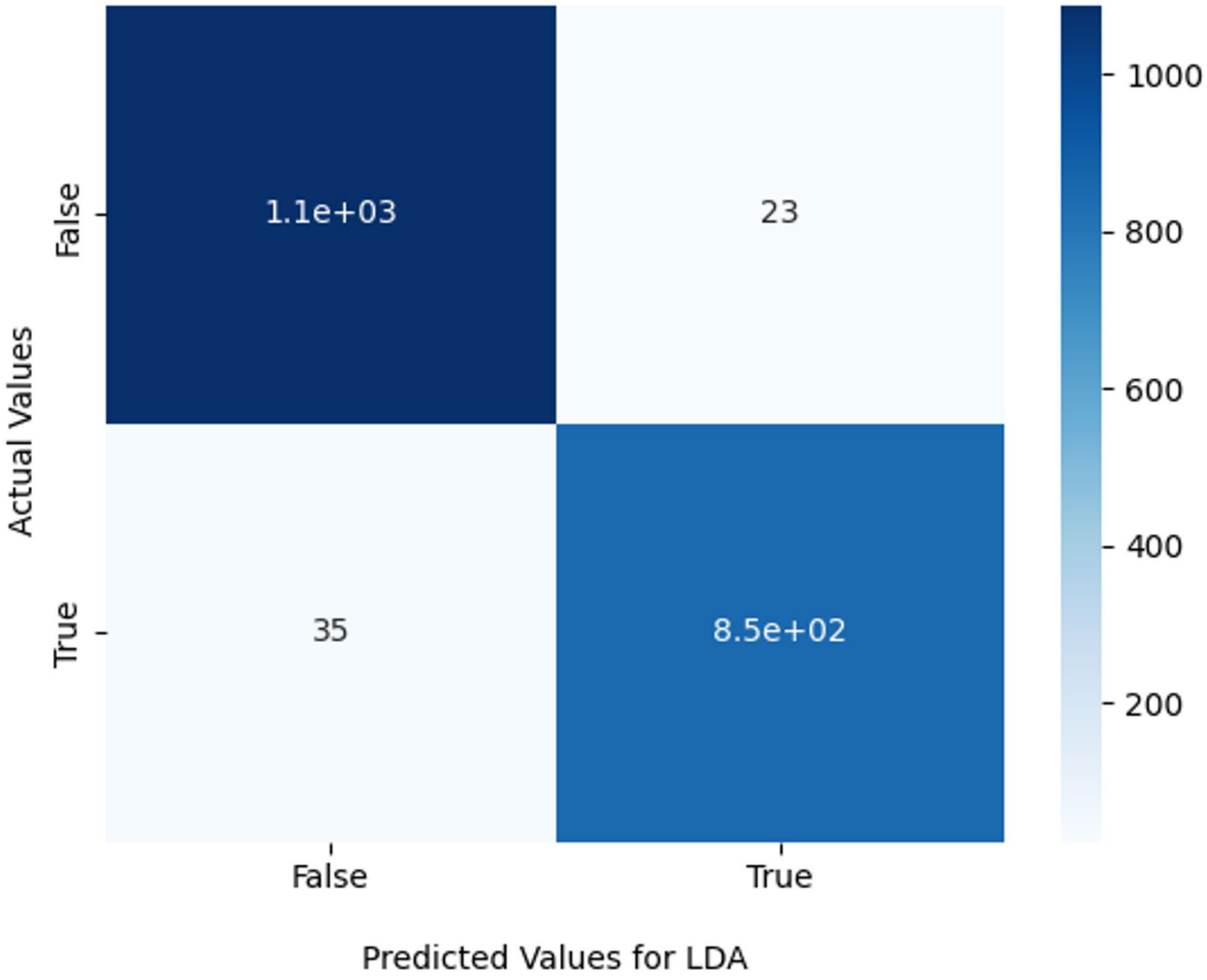

Fig. 5. Confusion matrix for LDA model.

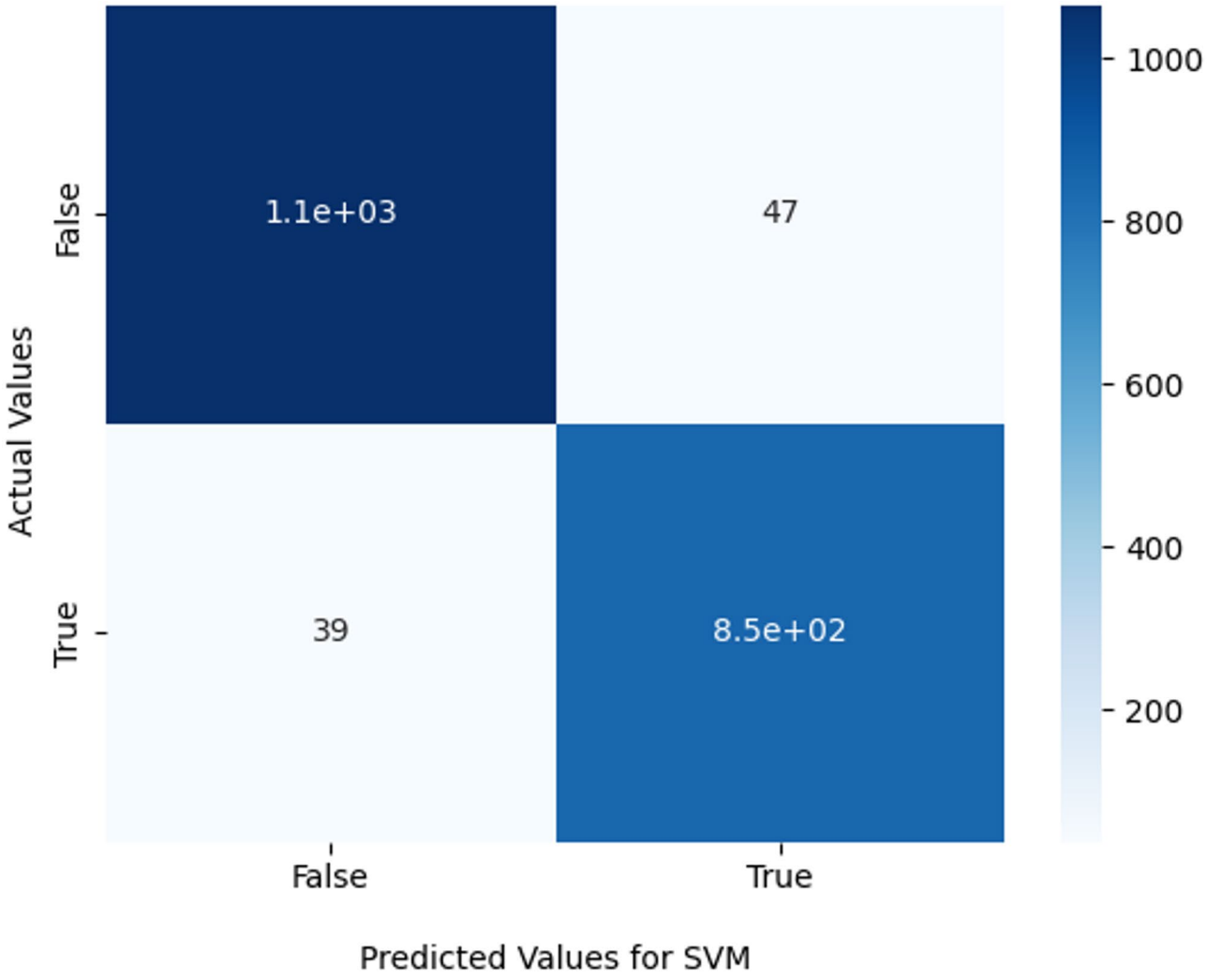

Fig. 6. Confusion matrix for SVM model.

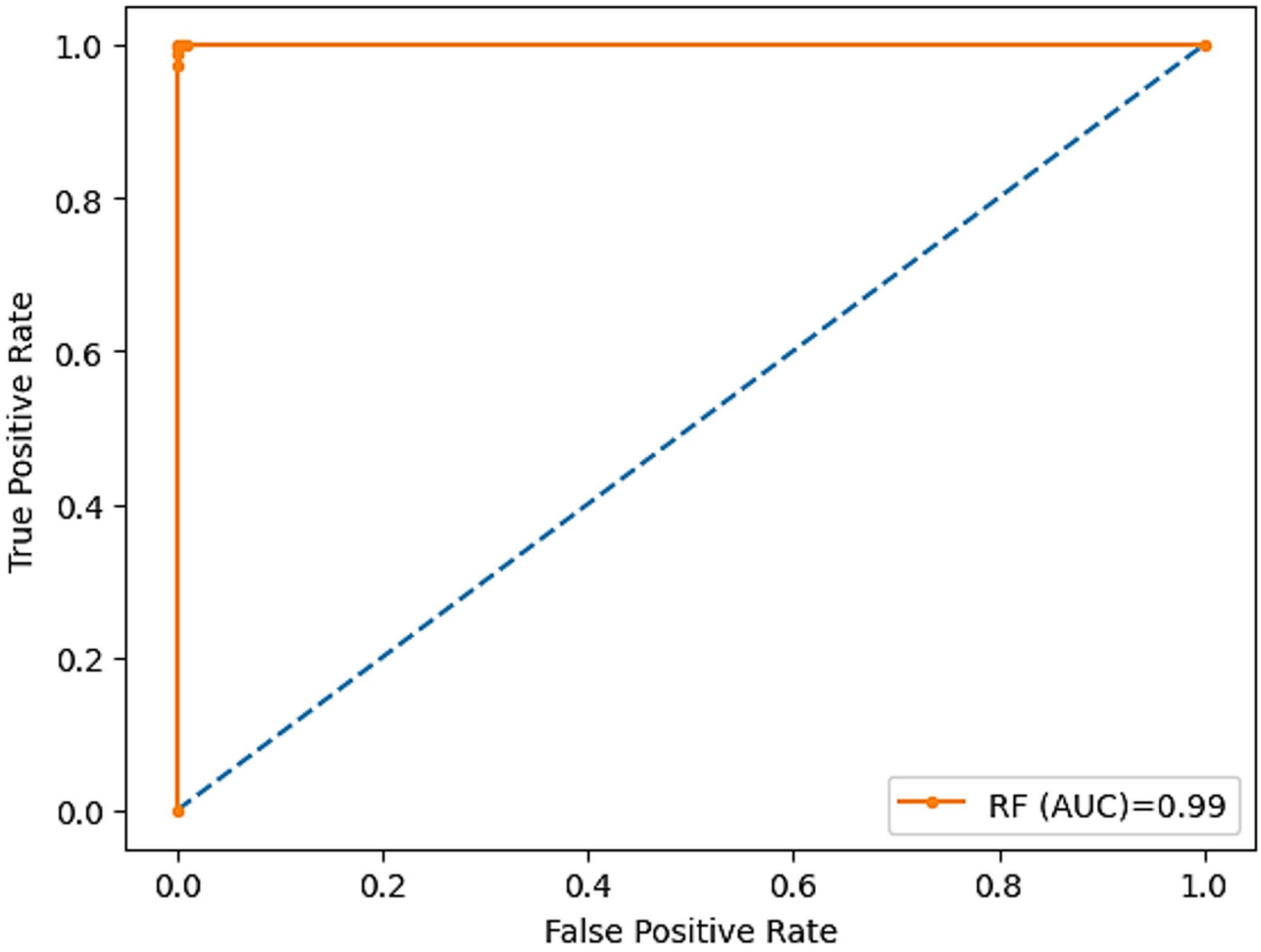

Fig. 7. ROC curve for RF model.

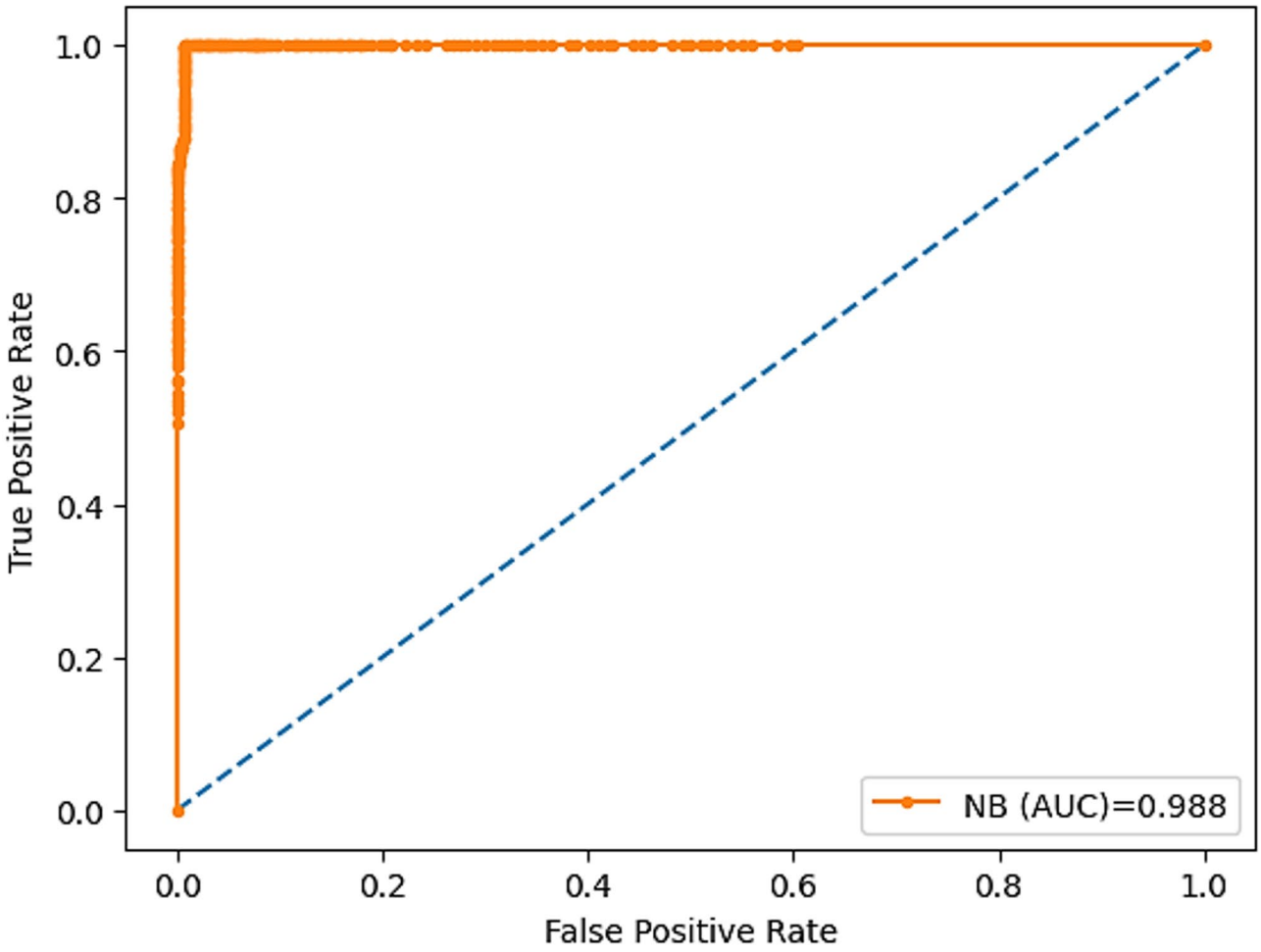

Fig. 8. ROC curve for NB model.

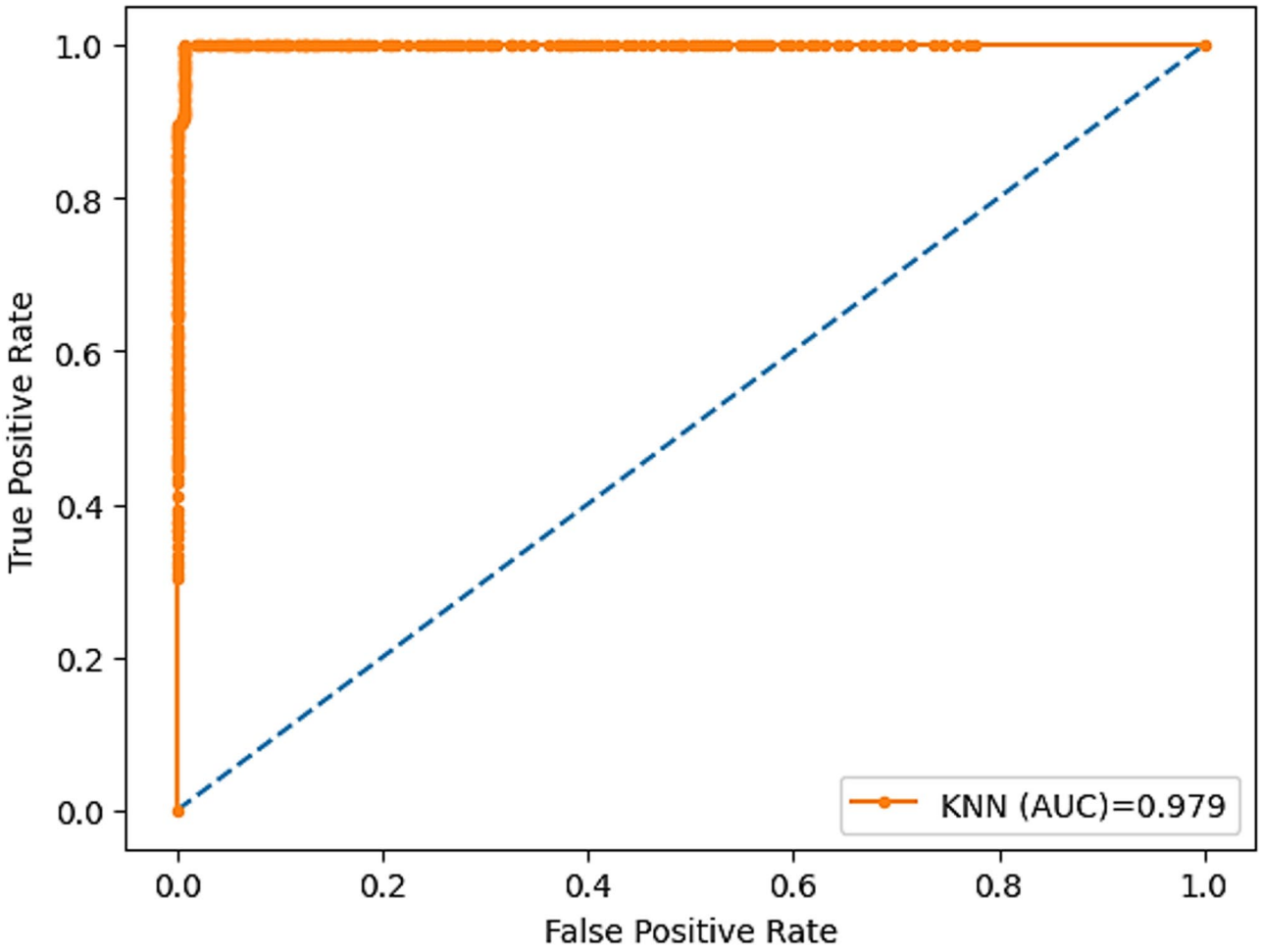

Fig. 9. ROC curve for KNN model.

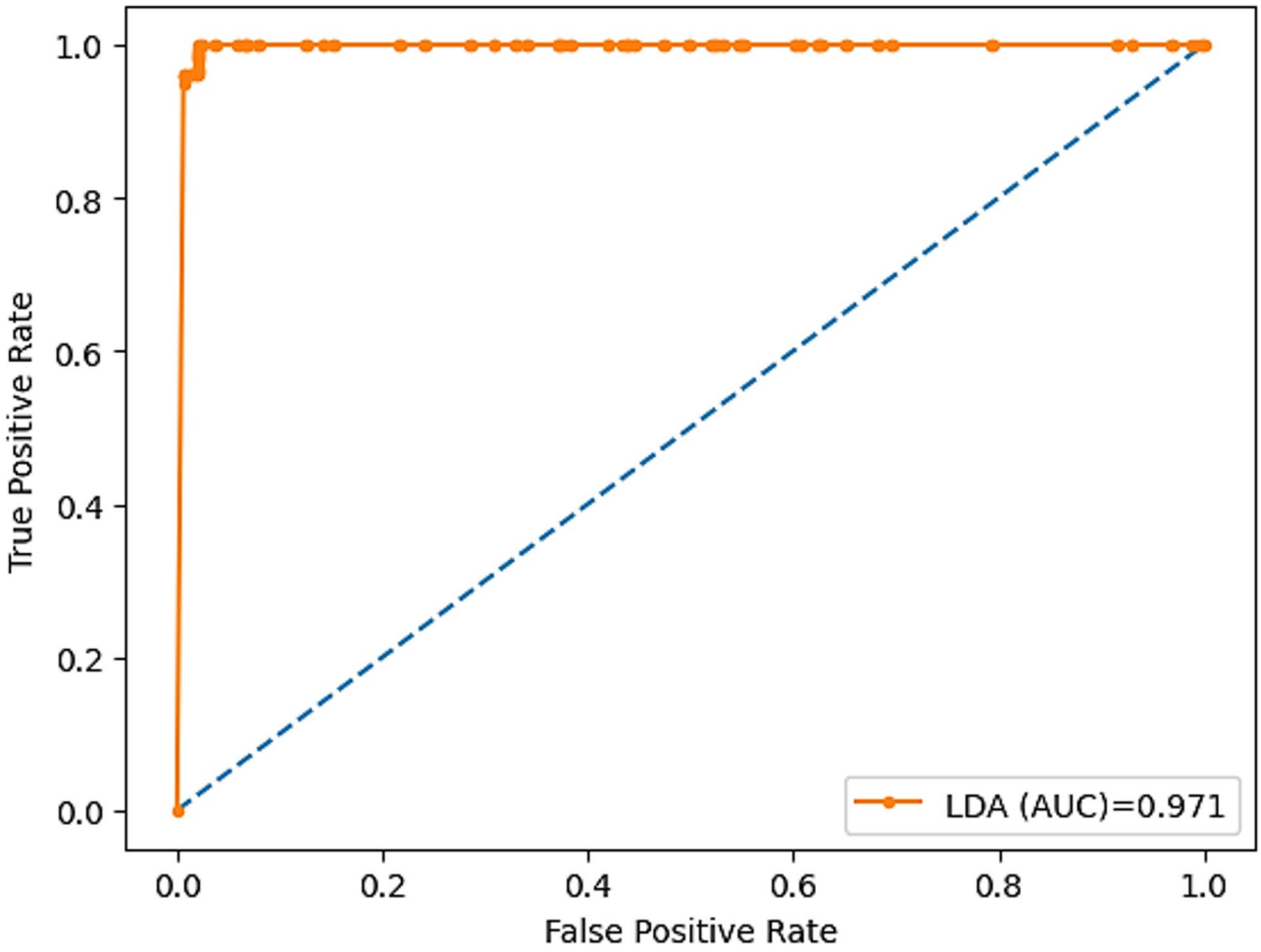

Fig. 10. ROC curve for LDA model.

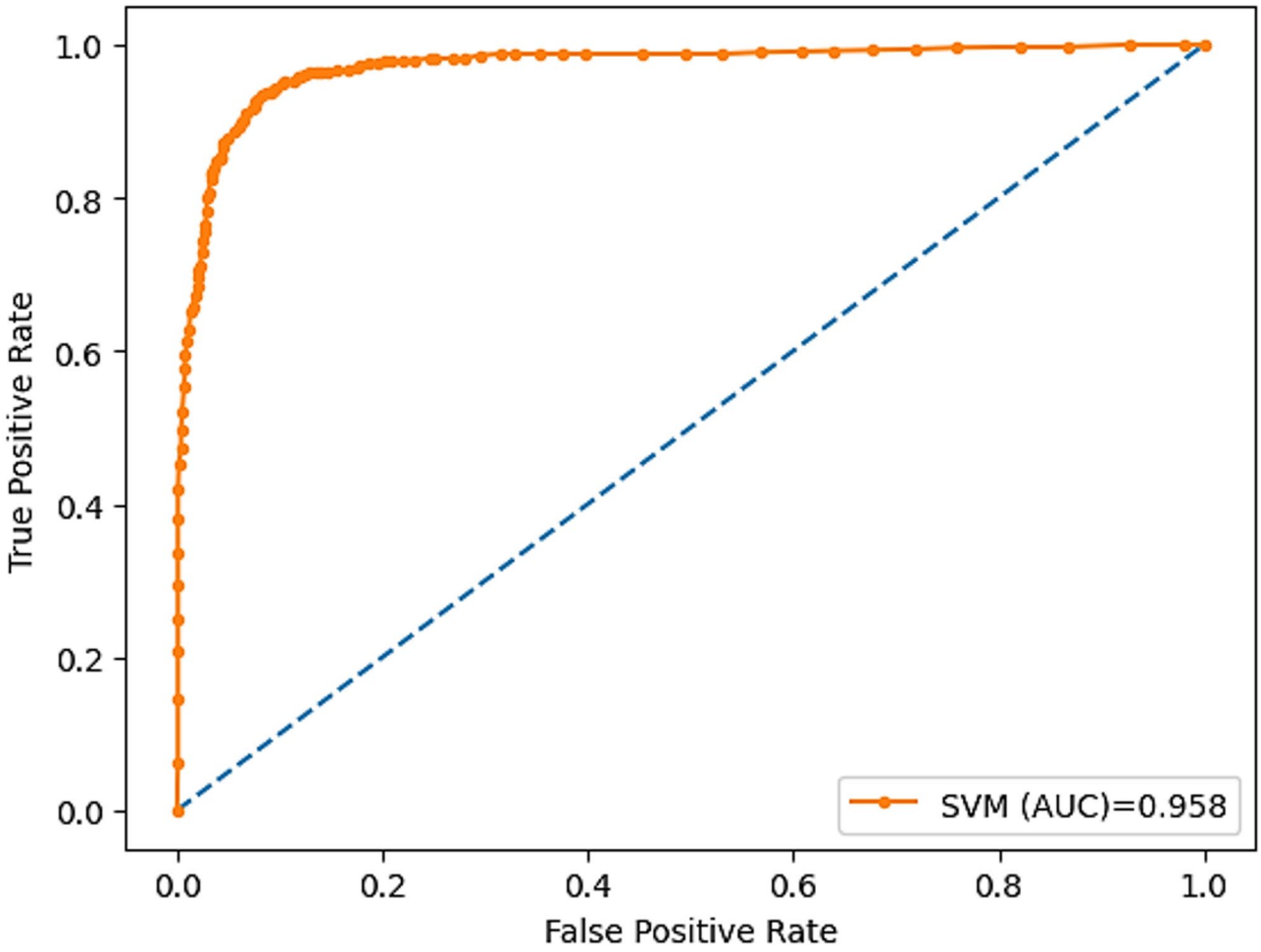

Fig. 11. ROC curve for SVM model.

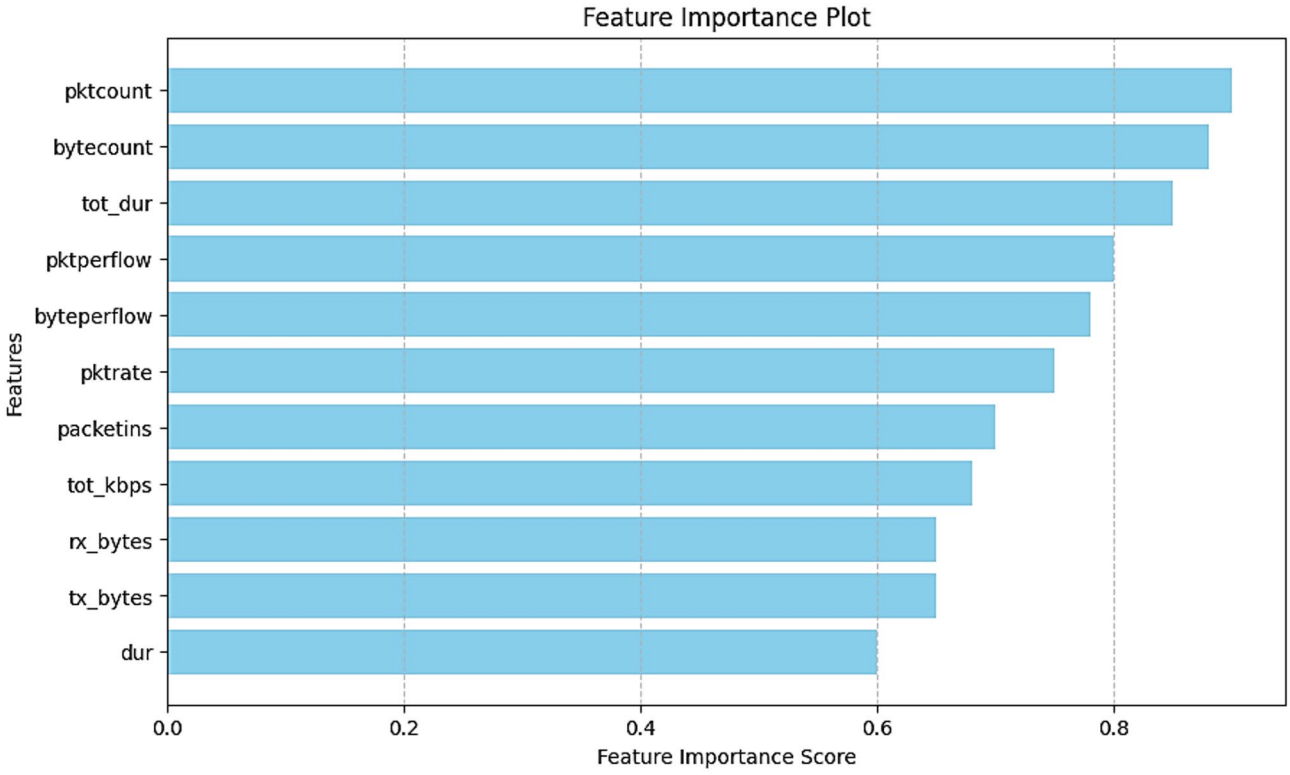

Fig. 12. Feature importance plot.

Models	Accuracy	F1 score	Recall	Precision
GS-RF	99.99%	99.99%	99.99%	99.99%
RS-RF	99.57%	99.57%	99.56%	99.57%

Table 5. Performance for the RF model using grid search (GS) and random search (RS).

Model	RF	NB	KNN	LDA	SVM
Theoretical Median	0.0	0.0	0.0	0.0	0.0
Actual Median	99.9	98.8	97.9	97.1	95.7
Number of Values	6	6	6	6	6
Sum of Signed Ranks (W)	0.0	0.0	0.0	0.0	0.0
Sum of Positive Ranks	0.0	0.0	0.0	0.0	0.0
Sum of Negative Ranks	0.0	0.0	0.0	0.0	0.0
P Value (Two-Tailed)	0.003125	0.003125	0.003125	0.003125	0.003125
Exact or Estimate?	Exact	Exact	Exact	Exact	Exact
P Value Summary	**	**	**	**	**
Significant (alpha = 0.05)?	Yes	Yes	Yes	Yes	Yes
Discrepancy	99.9	97.8	97.9	97.1	95.7

Table 6. Wilcoxon rank sum test results for RF model and another models.

Data availability

This study used DDoS SDN dataset, available at https://www.kaggle.com/datasets/aikenkazin/ddos-sdn-dataset that contains 104345 rows and 23 columns. There is a one target variable called label: contains only 1(malicious) and 0(benign).

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References

1. Sangodoyin, A. O., Akinsolu, M. O., Pillai, P. & Grout, V. Detection and classification of DDoS flooding attacks on software-defined networks: A case study for the application of machine learning. *IEEE Access* **9**, 122495–122508 (2021).
2. Jain, S. et al. B4: Experience with a globally-deployed software-defined WAN. *ACM SIGCOMM Comput. Commun. Rev.* **43**(4), 3–14 (2013).
3. Akgun, D., Hizal, S. & Cavusoglu, U. A new DDoS attacks intrusion detection model based on deep learning for cybersecurity. *Comput. Secur.* **118**, 102748 (2022).
4. McKeown, N. et al. OpenFlow: Enabling innovation in campus networks. *ACM SIGCOMM Comput. Commun. Rev.* **38**(2), 69–74 (2008).
5. Sarica, A. K. & Angin, P. Explainable security in SDN-based IoT networks. *Sensors* **20**(24), 7326 (2020).
6. Kreutz, D. et al. Software-defined networking: A comprehensive survey. *Proc. IEEE* **103**(1), 14–76 (2015).
7. Sangodoyin, A., Sigwele, T., Pillai, P., Hu, Y. F., Awan, I. & Disso, J. DoS attack impact assessment on software-defined networks. In *Proceedings of International Conference Wireless Satellite System* 11–22 (Springer, Cham, 2017).
8. Sezer, S. et al. Are we ready for SDN? Implementation challenges for software-defined networks. *IEEE Commun. Mag.* **51**(7), 36–43 (2013).
9. Almaraz-Rivera, J. G., Perez-Diaz, J. A. & Cantoral-Ceballos, J. A. Transport and application layer DDoS attacks detection to IoT devices by using machine learning and deep learning models. *Sensors* **22**(9), 3367 (2022).
10. Bhattacharyya, D. K. & Kalita, J. K. *DDoS Attacks: Evolution, Detection, Prevention, Reaction, and Tolerance* (CRC Press, 2016).
11. Cirillo, M., Mauro, M. D., Matta, V. & Tambasco, M. Botnet identi_cation in DDoS attacks with multiple emulation dictionaries. *IEEE Trans. Inf. Forensics Secur.* **16**, 3554–3569 (2021).
12. Ko, I., Chambers, D. & Barrett, E. Feature dynamic deep learning approach for DDoS mitigation within the ISP domain. *Int. J. Inf. Secur.* **19**, 53–70 (2020).
13. Ortet Lopes, I., Zou, D., Ruambo, F. A., Akbar, S. & Yuan, B. Towards effective detection of recent DDoS attacks: A deep learning approach. *Secur. Commun. Netw.* **2021**(1), 5710028 (2021).
14. Nandhini, P. S., Kuppuswami, S., Abinaya, C., Boomika, R. B. & Deepthi, N. Classification and analysis of distributed denial of service attacks using machine learning techniques. In *Proceedings of 5th International Conference on Advances in Applied Science and Environmental Technology (ICAST)* 302–305 (IEEE, 2022).
15. Sharif, D. M. Application-layer DDoS detection via efficient machine learning and feature selection. In *Proceedings of International Conference on Engineering Applied and Nano Sciences. (ICEANS)* 19–23 (IEEE, 2023).
16. Ghurkan, D. K. & Abdulrahman, A. A. Construct an efficient DDoS attack detection system based on RF-C4. 5-GridSearchCV. In *2022 Iraqi International Conference on Communication and Information Technologies (IICCIT)* 120–124 (IEEE, 2022).
17. Haribalaji, S. & Ranjana, P. Distributed denial of service (DDoS) attack detection using classification algorithm. In *Proceedings of International Conference on Advances in Data Engineering and Intelligent Computing Systems (ADICS)* 1–6 (IEEE, 2024)
18. Elshabrawy, M. A review on waste management techniques for sustainable energy production. *Metaheuristic Optim. Rev.* **3**, 47–58 (2025).
19. El-Kenawy, E. S. M. et al. Greylag goose optimization: nature-inspired optimization algorithm. *Expert Syst. Appl.* **238**, 122147 (2024).
20. Khaled, K. & Singla, M. K. Predictive analysis of groundwater resources using random forest regression. *J. Artif. Intell. Metaheuristics* **09**, 11–19 (2025).
21. Mishra, P. et al. Forecasting production of potato for a sustainable future: global market analysis. *Potato Res.* **67**(4), 1671–1690 (2024).
22. Atteia, G. et al. Adaptive dynamic dipper throated optimization for feature selection in medical data. *Comput., Mater. Contin.* **75**(1), 1883–1900 (2023).
23. Vakili, A. et al. A new service composition method in the cloud-based internet of things environment using a grey wolf optimization algorithm and MapReduce framework. *Concurr. Comput. Pract. Exp.* **36**(16), e8091 (2024).
24. Amiri, Z., Heidari, A., Navimipour, N. J., Esmaeilpour, M. & Yazdani, Y. The deep learning applications in IoT-based bio-and medical informatics: A systematic literature review. *Neural Comput. Appl.* **36**(11), 5757–5797 (2024).
25. Heidari, A., Jafari Navimipour, N., Jabraeil Jamali, M. A. & Akbarpour, S. Securing and optimizing IoT offloading with blockchain and deep reinforcement learning in multi-user environments. *Wirel. Netw.* **31**, 1–22 (2025).
26. Heidari, A., Amiri, Z., Jamali, M. A. J. & Jafari, N. Assessment of reliability and availability of wireless sensor networks in industrial applications by considering permanent faults. *Concurr. Comput. Pract. Exp.* **36**(27), e8252 (2024).
27. Amiri, Z., Heidari, A., Zavvar, M., Navimipour, N. J. & Esmaeilpour, M. The applications of nature-inspired algorithms in Internet of Things-based healthcare service: A systematic literature review. *Trans. Emerg. Telecommun. Technol.* **35**(6), e4969 (2024).
28. Zanbouri, K. et al. A GSO-based multi-objective technique for performance optimization of blockchain-based industrial Internet of things. *Int. J. Commun. Syst.* **37**(15), e5886 (2024).
29. Heidari, A., Navimipour, N. J., Zeadally, S. & Chamola, V. Everything you wanted to know about ChatGPT: Components, capabilities, applications, and opportunities. *Internet Technol. Lett.* **7**(6), e530 (2024).
30. Amiri, Z., Heidari, A. & Navimipour, N. J. Comprehensive survey of artificial intelligence techniques and strategies for climate change mitigation. *Energy* **308**, 132827 (2024).
31. Asadi, M., Jamali, M. A. J., Heidari, A. & Navimipour, N. J. Botnets unveiled: A comprehensive survey on evolving threats and defense strategies. *Trans. Emerg. Telecommun. Technol.* **35**(11), e5056 (2024).
32. Heidari, A., Amiri, Z., Jamali, M. A. J. & Navimipour, N. J. Enhancing solar convection analysis with multi-core processors and GPUs. *Eng. Rep.* **7**(1), e13050 (2025).
33. Wang, Y. et al. A nano-scale design of Vedic multiplier for electrocardiogram signal processing based on a quantum technology. *APL Mater.* **13**(3), 031114 (2025).
34. Chu, T. S., Si, W., Simoff, S. & Nguyen, Q. V. A machine learning classification model using random forest for detecting DDoS attacks. In *Proceedings of International Symposium on Networks, Computers and Communications (ISNCC)*, 1–7 (IEEE, 2022).
35. Dasari, K. B. & Devarakonda, N. Detection of DDoS attacks using machine learning classification algorithms. *Int. J. Comput. Netw. Inf. Secur.* **9**(6), 89 (2022).
36. Faruq, Z. F., Mantoro, T., Bhakti, M. A. C. Random forest classifier evaluation in DDoS detection system for cyber defence preparation. In *Proceedings of 8th International Conference on Computing, Engineering and Design (ICCED)*, 1–5 (IEEE, 2022).

37. Bhaskaraa, I. M. et al. Klasifikasi Serangan distributed denial of service (DDoS) Menggunakan random forest dengan CFS. *J. Elektron. Ilmu Komput. Udayana* **11**, 215 (2022).
38. Singh Samom, P. & Taggu, A. Distributed denial of service (DDoS) attacks detection: A machine learning approach. In *Applied Soft Computing and Communication Networks: Proceedings of ACN 2020*, 75–87 (Springer, Singapore, 2021).
39. Mohmand, M. I. et al. A machine learning-based classification and prediction technique for DDoS attacks. *IEEE Access* **10**, 21443–21454 (2022).
40. Ivanova, V., Tashev, T. & Draganov, I. Random forest detector and classifier of multiple IoT-based DDoS attacks. *WSEAS Trans. Inf. Sci. Appl.* **19**, 30–43 (2022).
41. Maslan, A., Mohamad, K. M. B. & Foozy, F. B. M. Feature selection for DDoS detection using classification machine learning techniques. *IAES Int. J. Artif. Intell.* **9**(1), 137 (2020).
42. Robinson, R. R. & Thomas, C. Ranking of machine learning algorithms based on the performance in classifying DDoS attacks. In *Proceedings of IEEE Recent Advances in Intelligent Computational Systems (RAICS)*, 185–190 (IEEE, 2015).
43. Alkasassbeh, M., Al-Naymat, G., Hassanat, A. B. & Almseidin, M. Detecting distributed denial of service attacks using data mining techniques. *Int. J. Adv. Comput. Sci. Appl.* **7**(1), 436 (2016).
44. Fathima, A., Devi, G. S. & Faizaanuddin, M. Improving distributed denial of service attack detection using supervised machine learning. *Meas. Sens.* **30**, 100911 (2023).
45. Alduailij, M. et al. Machine-learning-based DDoS attack detection using mutual information and random forest feature importance method. *Symmetry* **14**(6), 1095 (2022).
46. Anyanwu, G. O., Nwakanma, C. I., Lee, J. M. & Kim, D. S. Optimization of RBF-SVM kernel using grid search algorithm for DDoS attack detection in SDN-based VANET. *IEEE Internet Things J.* **10**(10), 8477–8490 (2022).
47. Wei, Y. et al. AE-MLP: A hybrid deep learning approach for DDoS detection and classification. *IEEE Access* **9**, 146810–146821 (2021).
48. Habib, B. & Khursheed, F. Performance evaluation of machine learning models for distributed denial of service attack detection using improved feature selection and hyper-parameter optimization techniques. *Concurr. Comput. Pract. Exp.* **34**(26), e7299 (2022).
49. Amiri, Z., Heidari, A., Jafari, N. & Hosseinzadeh, M. Deep study on autonomous learning techniques for complex pattern recognition in interconnected information systems. *Comput. Sci. Rev.* **54**, 100666 (2024).
50. Jabraeil Jamali, M. A., Bahrami, B., Heidari, A., Allahverdizadeh, P., Norouzi, F., Jabraeil Jamali, M. A., Bahrami, B., Heidari, A., Allahverdizadeh, P. & Norouzi, F. The IoT Landscape. Towards the Internet of Things: Architectures, Security, and Applications 1–8 (2020).
51. Heidari, A., Jamali, M. A. J. & Navimipour, N. J. An innovative performance assessment method for increasing the efficiency of AODV routing protocol in VANETs through colored timed petri nets. *Concurr. Comput. Pract. Exp.* **37**(3), e8349 (2025).
52. Ahmadpour, S. S. et al. A nano-design of a quantum-based arithmetic and logic unit for enhancing the efficiency of the future IoT applications. *AIP Adv.* **15**(3), 035341 (2025).
53. Maleki, F., Jamali, M. A. J. & Heidari, A. Unmanned aerial vehicle routing based on frog-leaping optimization algorithm. *Sci. Rep.* **15**(1), 11249 (2025).
54. Elshewey, A. M. et al. Prediction of aerodynamic coefficients based on machine learning models. *Model. Earth Syst. Environ.* **11**(3), 184 (2025).
55. Fallatah, A., Massa'deh, M. O. & Alkouri, A. U. NORMAL AND COSETS OF (γ,)-FUZZY HX-SUBGROUPS. *J. Appl. Math. Inform.* **40**(3–4), 719–727 (2022).
56. Alkouri, A., AbuHijleh, E. A., Alafifi, G., Almuhur, E. & Al-Zubi, F. M. More on complex hesitant fuzzy graphs. *AIMS Math.* **8**(12), 30429–30444 (2023).
57. Alkhasawneh, M. S. Hybrid cascade forward neural network with Elman neural network for disease prediction. *Arab. J. Sci. Eng.* **44**(11), 9209–9220 (2019).

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The authors confirm contribution to the paper as follows: study conception and design: Hela Elmannai and Alaa A. El-Bary; data collection: Alaa A. El-Bary and Mohamed S. Sawah; analysis and interpretation of results: Mohamed S.Sawah and Osama E. Sheta; draft manuscript preparation: Osama E. Sheta, Kh. Lotfy, and Hela Elmannai.All authors reviewed the results and approved the final version of the manuscript.

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Declarations

Competing interests
The authors declare no competing interests.

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